

Skill - 6

1. Process Escape Sequences and Create Child Process

```
harshika@siri: ~$ gedit skill6.c

** (gedit:1713): WARNING **: 03:58:14.970: Could not load Peas repository: TypeLib file for namespace 'Peas', version '1.0' not found

** (gedit:1713): WARNING **: 03:58:14.970: Could not load PeasGtk repository : TypeLib file for namespace 'PeasGtk', version '1.0' not found
harshika@siri: ~$ gcc skill6.c -o skill6
harshika@siri: ~$ ./skill6

=====
SKILL 6 - PROCESS HANDLING
=====

PART A: PROCESS ESCAPE SEQUENCES
-----

Enter a command with escaped spaces/symbols.
Example: echo Hello\ World
Example: echo Hello\@World

Enter command: echo Hello\ World

----- PARSER OUTPUT -----
Number of arguments: 1
Argument 1: [echo]
-----

PART B: CHILD PROCESS CREATION
-----

Creating child process...

[PARENT PROCESS]
Parent PID : 2133
Child PID  : 2136
Waiting for child process...

[CHILD PROCESS]
Child PID   : 2136
Parent PID  : 2133
Executing command: echo

1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <string.h>
4 #include <unistd.h>
5 #include <sys/types.h>
6 #include <sys/wait.h>
7 #include <errno.h>
8
9 #define MAX_INPUT 200
10 #define MAX_ARGS 20
11
12
13 /* -----
14 PART A: COMMAND PARSER
15 ----- */
16
17 int parse_command(char *input, char *args[])
18 {
19     int count = 0;
20     char *p = input;
21
22     while (*p != '\0' && count < MAX_ARGS - 1)
23     {
24         /* Skip spaces */
25         while (*p == ' ')
26         {
27             p++;
28         }
29
30         if (*p == '\0')
31         {
32             break;
33         }
34
35         args[count] = p;
36         count++;
37
38         char *write = p;
39         int escaped = 0;
40
41         while (*p != '\0')
42         {
43             if (escaped)
44             {
45                 /*
46                  * Preserve the character after backslash.
47                  * Example:
48                  * Hello\ World
49                  * becomes
50                  * Hello World
51                  */
52                 *write++ = *p;
53                 escaped = 0;
54             }
55             else if (*p == '\\')
```